

The exact upper logarithmic density of monochromatic subset sums

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Machine-generated note

17 September 2026

Abstract

For $r \geq 2$, let c_r be the least possible largest upper logarithmic density of a monochromatic finite-subset-sum set in an r -colouring of the positive integers. If $b > 1$ is the unique root of $b^r - 2rb + r - 1 = 0$, we prove

$$c_r = \frac{b^{r-1}}{2r}.$$

This proves Conlon–Fox–Pham’s tightness conjecture. The new step is a geometric covering lemma valid for arbitrary non-cyclic colour words; the additive reduction and matching construction are included self-containedly but are attributed to the stronger published results of Conlon, Fox and Pham.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 4](#) records the full provenance.

1 Introduction

For $A \subseteq \mathbb{N}$, write

$$\Sigma(A) = \left\{ \sum_{a \in F} a : \emptyset \neq F \subseteq A, F \text{ finite} \right\}, \quad \bar{d}_{\log}(S) = \limsup_{x \rightarrow \infty} \frac{1}{\log x} \sum_{\substack{n \leq x \\ n \in S}} \frac{1}{n},$$

and define

$$c_r = \inf_{\mathbb{N} = A_1 \cup \dots \cup A_r} \max_{i \in [r]} \bar{d}_{\log}(\Sigma(A_i)).$$

Conlon, Fox and Pham [1] proved the matching upper bound and the case $r = 2$. Their concluding conjecture is:

Conjecture 1.1 ([1, Concluding remarks]). We suspect that our upper bound is also tight for all $r \geq 3$, but our methods do not seem sufficient for proving this.

The catalog rendering attached a formula with $2^r b - r$ in the denominator. The source paper has $2rb - r$. This distinction is invisible when $r = 2$ and substantive thereafter; [Remark 3.5](#) records it.

Theorem 1.2 (Main theorem). *Let $r \geq 2$, and let $b > 1$ be the unique root of*

$$b^r - 2rb + r - 1 = 0.$$

Then

$$c_r = \frac{b^{r-1}}{2r} = \frac{1 - \frac{1}{2b}}{1 - b^{-r}} = \left(1 - \frac{1}{2b}\right) \left(1 + \frac{1}{2rb - r}\right).$$

Idea of proof. At every dyadic scale one colour contains a positive proportion of the primes. A published additive lemma of Conlon–Fox–Pham, or the weaker self-contained prime-block lemma below, makes that colour's subset sums contain an interval with logarithmic endpoints $\log n + O(\log \log n)$ and $2 \log n - O(\log \log n)$. The central geometric lemma proves that arbitrarily interlaced intervals $[t_k, \lambda t_{k+1}]$ force one of r colours to have upper density at least $\inf_{q>1} (1 - 1/(\lambda q))/(1 - q^{-r})$. Letting $\lambda \uparrow 2$ yields the lower bound. The block construction of [1, Theorem 1] gives equality. [\[full proof\]](#)

2 The geometric step

For $r \geq 2, \lambda > 1$ put

$$D_{r,\lambda}(q) = \frac{1 - \frac{1}{\lambda q}}{1 - q^{-r}}, \quad C_{r,\lambda} = \inf_{q>1} D_{r,\lambda}(q).$$

Lemma 2.1 (Geometric covering lemma). *Let $0 < t_0 < t_1 < \dots$ tend to infinity, label each k by $\gamma_k \in [r]$, and set*

$$J_k = [t_k, \lambda t_{k+1}], \quad U_i = \bigcup_{\gamma_k=i} J_k.$$

Then

$$\max_{i \in [r]} \limsup_{T \rightarrow \infty} \frac{|U_i \cap [0, T]|}{T} \geq C_{r,\lambda}.$$

Idea of proof. Assume every density is below $D < C_{r,\lambda}$ and decompose each U_i into interval components. For a component C , its normalized length a_C , endpoint density d_C , span ratio Q_C , and ratio R_C to the preceding same-colour endpoint satisfy $d_C = a_C + d_{C-}/R_C$. The definition of $C_{r,\lambda}$ makes this recurrence lose a uniform amount $\eta > 0$ per component after taking logs. Summing over colours, the endpoint and span terms cancel up to $O(1)$, while $-\eta$ times the number of components tends to $-\infty$. [\[full proof\]](#)

Lemma 2.2 (Evaluation of the constant). *The polynomial $q^r - \lambda r q + r - 1$ has a unique root $b_{r,\lambda} > 1$, which is the minimizer of $D_{r,\lambda}$, and*

$$C_{r,\lambda} = \frac{b_{r,\lambda}^{r-1}}{\lambda r}.$$

Consequently $C_{r,\lambda} \rightarrow b^{r-1}/(2r)$ as $\lambda \uparrow 2$.

Idea of proof. Differentiate $D_{r,\lambda}$; its critical equation is the displayed polynomial, which decreases then increases and has one root beyond 1. At the root, $b_{r,\lambda}^r - 1 = r(\lambda b_{r,\lambda} - 1)$, which reduces the value to the stated form. [\[full proof\]](#)

3 Additive input and transfer

Lemma 3.1 (Restricted sums modulo a prime). *Let p be prime and $B \subseteq \mathbb{F}_p$ have size s . If $1 \leq k \leq s$ and $k(s - k) \geq p - 1$, every element of $\mathbb{F}_p$ is a sum of k distinct members of B .*

Idea of proof. This is the Dias da Silva–Hamidoune theorem [3], with the polynomial proof of Alon–Nathanson–Ruzsa [4]. For completeness, apply the coefficient form of the Combinatorial Nullstellensatz to

$$((x_1 + \dots + x_k - c)^{p-1} - 1) \prod_{i < j} (x_j - x_i).$$

The omitted determinant calculation in the generated writeup is supplied in the full proof using falling factorials. [\[full proof\]](#)

Lemma 3.2 (Prime-block interval lemma). *For every fixed $\kappa > 0$ and all sufficiently large n , if A is a set of primes in $[n, 2n]$ with $|A| \geq \kappa n / \log n$, then*

$$\left[n(\log n)^3, \frac{n^2}{(\log n)^3} \right] \cap \mathbb{Z} \subseteq \Sigma(A).$$

Idea of proof. Partition A into a group-building half and a reserve half. Repeated use of [Lemma 3.1](#) modulo a prime $p \in (2n, 4n)$ produces many disjoint equal-sum groups, giving a long arithmetic progression of subset sums. The reserve represents every residue modulo the common difference, by another use of [Lemma 3.1](#) and the Chinese remainder theorem. Adding the two representations fills the interval. This lemma is weaker than [\[1, Lemma 2\]](#), which already supplies the additive input needed here. [\[full proof\]](#)

Remark 3.3 (What is new). The additive interval step, its transfer from dyadic scales to logarithmic density, and the upper construction were already proved in stronger form in [\[1\]](#), using the companion paper [\[2\]](#). In [\[1, Lemma 7\]](#), the authors proved the geometric statement for two colours and noted that their method extends to cyclic auxiliary colourings. The genuinely new claim here is [Lemma 2.1](#) for arbitrary non-cyclic colour words. The referee identified this correctly; the generated writeup’s description of the prime lemma as the key new step was wrong.

Remark 3.4 (Computational audit). The referee verified all constant identities symbolically; tested [Lemma 3.1](#) on 7,740 cases with no failure; audited all telescoping and covering identities of [Lemma 2.1](#) on 22 configurations up to 128,000 blocks; and exhaustively searched all colour words up to rotation and colour permutation for $r = 2$ through period 8, $r = 3$ through period 7, and $r = 4, 5$ through period 6. No configuration fell below C_r . These checks are evidence, not part of the proof.

Remark 3.5 (Constant transcription). The source equation gives $b^r - 1 = 2rb - r$, so the denominator is $2rb - r$. For $r \geq 3$, $2^r > 2r$; hence the catalog expression using $2^r b - r$ is strictly smaller than the correct value and is false even as an upper bound.

4 Provenance

The mathematics was produced without human intervention in three stages. (1) The result was found by gpt-6-astra, reasoning effort max, in one pass begun on 2026-09-17, for catalog id 2105.15195__00 in attacks_retry. The prompt included a gpt-5.6-sol attempt. Its geometric lemma and telescoping skeleton were inherited; the retry tightened the boundary accounting and added a prime-block argument. (2) An adversarial claudes-opus-5 referee re-derived the argument, checked the source and companion papers, and ran the computations above on 2026-09-17; its verdict was MINOR GAPS, at medium confidence because the remaining risk is concentrated in the $O(1)$ bookkeeping of [Lemma 2.1](#). (3) On 2026-09-17 claudes-fable-5-1 created only a recoverable note skeleton before its session ended. On 2026-09-18 Codex completed the present rewrite, put full proofs in [Appendix A](#), and made these declared changes: the determinant identity omitted from the restricted-sum proof is proved explicitly; Dias da Silva–Hamidoune and Alon–Nathanson–Ruzsa are cited; the prime-block lemma, transfer, and construction are identified as published or weaker rediscoveries; the geometric lemma is identified as the new step; the source’s $2r$ constant is distinguished from the catalog’s 2^r transcription. No other mathematics was changed. **No human mathematician has checked this argument.**

A Full proofs

Lemma 2.1 (Geometric covering lemma). *Let $0 < t_0 < t_1 < \dots$ tend to infinity, label each k by $\gamma_k \in [r]$, and set*

$$J_k = [t_k, \lambda t_{k+1}], \quad U_i = \bigcup_{\gamma_k=i} J_k.$$

Then

$$\max_{i \in [r]} \limsup_{T \rightarrow \infty} \frac{|U_i \cap [0, T]|}{T} \geq C_{r, \lambda}.$$

Proof. Suppose the conclusion fails. Choose D with

$$\max_i \limsup_{T \rightarrow \infty} \frac{|U_i \cap [0, T]|}{T} < D < C_{r, \lambda}.$$

For all sufficiently large T , $|U_i \cap [0, T]| \leq DT$ for every i . No U_i has an unbounded component, since that would give density one. Each component is therefore an interval $C = [t_f, \lambda t_{\ell+1}]$. Put

$$e_C = \lambda t_{\ell+1}, \quad Q_C = \frac{t_{\ell+1}}{t_f}, \quad a_C = \frac{|C|}{e_C} = 1 - \frac{1}{\lambda Q_C}, \quad d_C = \frac{|U_i \cap [0, e_C]|}{e_C}.$$

If C^- is the preceding component of the same colour and $R_C = e_C/e_{C^-}$, disjointness gives

$$d_C = a_C + \frac{d_{C^-}}{R_C}. \quad (1)$$

Discard finitely many components so that each retained component has a predecessor and $d_C \leq D$. Then

$$1 - \frac{1}{\lambda} < a_C < d_C \leq D, \quad 1 < Q_C < Q_{\max} := \frac{1}{\lambda(1-D)}.$$

From (1),

$$\log \frac{d_{C^-}}{d_C} \leq \log R_C + \log \left(1 - \frac{a_C}{D}\right). \quad (2)$$

On $1 \leq q < Q_{\max}$ define

$$F(q) = r \log q + \log \left(1 - \frac{1 - \frac{1}{\lambda q}}{D}\right).$$

For $q > 1$, $F(q) \geq 0$ is equivalent to $D \geq D_{r, \lambda}(q)$, contrary to $D < C_{r, \lambda}$. Moreover $F(1) < 0$ and $F(q) \rightarrow -\infty$ as $q \uparrow Q_{\max}$. Continuity on the intervening compact range gives $\varepsilon > 0$ such that every retained component satisfies

$$\log \left(1 - \frac{a_C}{D}\right) \leq -r \log Q_C - \varepsilon. \quad (3)$$

Sum (2)–(3) over retained components with last index at most N , and call their number m_N . For each colour, the left side telescopes; since $1 - 1/\lambda \leq d_C \leq D$, its total is $O(1)$. Endpoint ratios also telescope, giving

$$\sum_C \log R_C \leq r \log t_{N+1} + O(1).$$

Finally,

$$\sum_C \log Q_C \geq \log t_{N+1} - O(1). \quad (4)$$

Indeed, $\log Q_C = \sum_{k=f(C)}^{\ell(C)} \log(t_{k+1}/t_k)$, and every index is in the span of its own colour component. Discarded components cost only a fixed initial amount. At the right boundary, at most one component of each colour straddles N , and each omitted span is at most $\log Q_{\max}$. This proves (4).

Combining the estimates gives

$$O(1) \leq r \log t_{N+1} - r \sum_C \log Q_C - \varepsilon m_N + O(1) \leq O(1) - \varepsilon m_N.$$

But $m_N \rightarrow \infty$: there are infinitely many intervals, every component is finite, and every finite component contains only finitely many intervals. This contradiction proves the lemma. $\square$

Lemma 2.2 (Evaluation of the constant). *The polynomial $q^r - \lambda r q + r - 1$ has a unique root $b_{r,\lambda} > 1$, which is the minimizer of $D_{r,\lambda}$, and*

$$C_{r,\lambda} = \frac{b_{r,\lambda}^{r-1}}{\lambda r}.$$

Consequently $C_{r,\lambda} \rightarrow b^{r-1}/(2r)$ as $\lambda \uparrow 2$.

Proof. Differentiation gives

$$D'_{r,\lambda}(q) = 0 \iff q^r - \lambda r q + r - 1 = 0.$$

The polynomial is negative at $q = 1$, decreases until $q = \lambda^{1/(r-1)}$, and then increases to infinity. It has a unique root $b_{r,\lambda} > 1$, necessarily beyond $\lambda^{1/(r-1)}$, and this is the unique minimizer. Since

$$b_{r,\lambda}^r - 1 = r(\lambda b_{r,\lambda} - 1),$$

direct substitution yields $C_{r,\lambda} = b_{r,\lambda}^{r-1}/(\lambda r)$. Continuity of the unique root as $\lambda \uparrow 2$ proves the limit. $\square$

Lemma 3.1 (Restricted sums modulo a prime). *Let p be prime and $B \subseteq \mathbb{F}_p$ have size s . If $1 \leq k \leq s$ and $k(s-k) \geq p-1$, every element of $\mathbb{F}_p$ is a sum of k distinct members of B .*

Proof. Suppose $c \in \mathbb{F}_p$ is not a sum of k distinct members of B . Then

$$P(x_1, \dots, x_k) = ((x_1 + \dots + x_k - c)^{p-1} - 1) \prod_{i < j} (x_j - x_i)$$

vanishes on B^k : repeated coordinates kill the Vandermonde factor, and otherwise Fermat's theorem kills the first factor.

Choose distinct integers $0 \leq d_1 < \dots < d_k \leq s-1$ with

$$\sum_i d_i = p-1 + \binom{k}{2}.$$

Such a choice exists because the sums of k distinct elements of $\{0, \dots, s-1\}$ fill $[\binom{k}{2}, \binom{k}{2} + k(s-k)]$. Explicitly, if $p-1 = kq + u$, $0 \leq u < k$, use $d_i = i-1 + q$ for the first $k-u$ indices and $d_i = i + q$ for the last u .

The coefficient of $\prod_i x_i^{d_i}$ in P is

$$\frac{(p-1)!}{\prod_i d_i!} \prod_{i < j} (d_j - d_i). \tag{5}$$

Here is the determinant step omitted from the generated writeup. Expanding the Vandermonde and the multinomial term gives, apart from $(p-1)!$,

$$\det \left(\frac{1}{(d_i - j + 1)!} \right)_{i,j=1}^k.$$

Multiply row i by $d_i!$. The j th entry becomes the falling factorial $(d_i)_{j-1}$, a monic polynomial of degree $j-1$ in d_i . Its determinant is therefore the Vandermonde $\prod_{i < j} (d_j - d_i)$. Dividing the row factors gives (5). This is the explicit MINOR_GAPS repair. The coefficient is nonzero in $\mathbb{F}_p$, because $d_i < p$ and the d_i are distinct.

Finally, the coefficient form of multivariate interpolation says that a polynomial of total degree $\sum_i d_i$ with this coefficient nonzero cannot vanish on a product of sets of sizes exceeding d_i . To see it, choose $X_i \subseteq B$ of size $d_i + 1$; the coefficient equals

$$\sum_{x_i \in X_i} \frac{P(x_1, \dots, x_k)}{\prod_i \prod_{y \in X_i \setminus \{x_i\}} (x_i - y)}.$$

This would vanish if P vanished on B^k , a contradiction. $\square$

Lemma 3.2 (Prime-block interval lemma). *For every fixed $\kappa > 0$ and all sufficiently large n , if A is a set of primes in $[n, 2n]$ with $|A| \geq \kappa n / \log n$, then*

$$\left[n(\log n)^3, \frac{n^2}{(\log n)^3} \right] \cap \mathbb{Z} \subseteq \Sigma(A).$$

Proof. Put $m = |A|$, choose by Bertrand's postulate a prime $2n < p < 4n$, and split A into a group-building set $A^{(0)}$ of size $\lfloor m/2 \rfloor$ and a reserve $A^{(1)}$. Let

$$h = \left\lceil \frac{8p}{m} \right\rceil = O_\kappa(\log n).$$

For large n , $h \leq m/8$. While at least $m/4$ elements of $A^{(0)}$ remain, [Lemma 3.1](#) supplies an h -set summing to zero modulo p , since for remaining size v , $h(v - h) \geq hm/8 \geq p$. This gives at least $m/(8h)$ disjoint groups. Each group sum is ℓp for some $1 \leq \ell \leq h$, so for one ℓ at least

$$L \geq \frac{m}{8h^2}$$

groups have common sum $d = \ell p$. Their subset sums contain $0, d, \dots, Ld$, whose upper endpoint $T = Ld$ satisfies

$$T \geq \frac{mn}{8h} \gg_\kappa \frac{n^2}{(\log n)^2}.$$

All reserve primes exceed ℓ . Choose a residue $c \bmod \ell$ for which

$$B = \{a \in A^{(1)} : a \equiv c \pmod{\ell}\} \quad \text{has} \quad s \geq \frac{m}{2\ell} \geq \frac{m}{2h}.$$

Thus $\gcd(c, \ell) = 1$. Set $H = \lceil 2p/s \rceil = O_\kappa((\log n)^2)$. For large n , $H + \ell \leq s/2$, and for each $t = H, \dots, H + \ell - 1$ we have $t(s - t) \geq p$. By [Lemma 3.1](#), the t -term sums from B cover every residue modulo p ; their residue modulo ℓ is tc . These ℓ choices of t cover all residues modulo ℓ . Since $\gcd(p, \ell) = 1$, the Chinese remainder theorem shows that every residue modulo $d = \ell p$ has a representative in $\Sigma(B)$ of size at most

$$M = 2n(H + \ell) = O_\kappa(n(\log n)^2).$$

For $M \leq x \leq T$, choose $b_x \leq M$ with $b_x \equiv x \pmod{d}$. Then $x - b_x = qd$ for some $0 \leq q \leq L$. Use q equal-sum groups together with the disjoint reserve representation. Hence $[M, T] \cap \mathbb{Z} \subseteq \Sigma(A)$. For large n , $M \leq n(\log n)^3$ and $T \geq n^2/(\log n)^3$, proving the claim. $\square$

Theorem 1.2 (Main theorem). *Let $r \geq 2$, and let $b > 1$ be the unique root of*

$$b^r - 2rb + r - 1 = 0.$$

Then

$$c_r = \frac{b^{r-1}}{2r} = \frac{1 - \frac{1}{2b}}{1 - b^{-r}} = \left(1 - \frac{1}{2b}\right) \left(1 + \frac{1}{2rb - r}\right).$$

Proof. Let $\mathbb{N} = A_1 \sqcup \dots \sqcup A_r$ be arbitrary and put $n_k = 2^k$. By the prime number theorem, $[n_k, 2n_k]$ contains at least $n_k/(2 \log n_k)$ primes for all sufficiently large k . Some colour γ_k contains at least $n_k/(2r \log n_k)$ of them. Applying [Lemma 3.2](#) with $\kappa = 1/(2r)$ gives

$$\left[n_k (\log n_k)^3, \frac{n_k^2}{(\log n_k)^3} \right] \cap \mathbb{Z} \subseteq \Sigma(A_{\gamma_k}). \quad (6)$$

As noted in the main text, the stronger published [[1](#), Lemma 2] can be used at this point instead.

Fix $1 < \lambda < 2$ and let $t_k = \log n_k + 3 \log \log n_k$. For all large k ,

$$\lambda t_{k+1} \leq 2 \log n_k - 3 \log \log n_k,$$

because the difference has leading term $(2 - \lambda)k \log 2$ and the remainder is $O_\lambda(\log k)$. Hence every integer whose logarithm is in $J_k = [t_k, \lambda t_{k+1}]$ belongs to $\Sigma(A_{\gamma_k})$.

For $E_i = \Sigma(A_i)$ define

$$U_i = \bigcup_{\gamma_k=i} J_k, \quad W_i = \bigcup_{n \in E_i} [\log n, \log(n+1)).$$

On each J_k , at most an initial interval of length e^{-t_k} is missing from W_i ; thus $|U_i \setminus W_i| < \infty$. Consequently

$$\begin{aligned} |U_i \cap [0, T]| &\leq |W_i \cap [0, T]| + O(1) \\ &\leq \sum_{\substack{n \leq e^T \\ n \in E_i}} \log \left(1 + \frac{1}{n} \right) + O(1) \leq \sum_{\substack{n \leq e^T \\ n \in E_i}} \frac{1}{n} + O(1). \end{aligned}$$

[Lemma 2.1](#) therefore yields

$$\max_i \bar{d}_{\log}(\Sigma(A_i)) \geq C_{r,\lambda}.$$

Letting $\lambda \uparrow 2$ and using [Lemma 2.2](#) gives

$$c_r \geq \frac{b^{r-1}}{2r}.$$

This transfer is the argument of [[1](#), §3 and Lemma 6]; it is included to show exactly how [Lemma 2.1](#) enters.

For the matching upper bound, use the construction of [[1](#), Theorem 1]. For all large k , let $N_k = \lceil e^{b^k} \rceil$ and colour $[N_k, N_{k+1})$ by $k \bmod r$. If a monochromatic subset sum has largest summand in block k , it lies in $[N_k, N_{k+1}^2]$. Apart from finitely many sums,

$$\Sigma(A_i) \subseteq \bigcup_{k \equiv i \pmod{r}} [N_k, N_{k+1}^2].$$

On the logarithmic scale, these intervals differ by summable endpoint errors from

$$V_i = \bigcup_{k \equiv i \pmod{r}} [b^k, 2b^{k+1}].$$

[Lemma 2.2](#) gives $b > 2^{1/(r-1)}$, so same-colour intervals are disjoint. At a right endpoint $2b^{k+1}$ their accumulated density tends to

$$\frac{\sum_{j \geq 0} (2b - 1)b^{k-rj}}{2b^{k+1}} = \frac{1 - \frac{1}{2b}}{1 - b^{-r}} = \frac{b^{r-1}}{2r}.$$

The density ratio increases within an occupied interval and decreases in a gap, so this is its limsup. Therefore $c_r \leq b^{r-1}/(2r)$, completing the proof. $\square$

B Report of the LLM referee (verbatim)

Status of this appendix

The following report is reproduced verbatim as an audit trail. Its references to the original writeup predate the determinant repair and the corrected scholarship framing in [Section 4](#).

B.1 Interpretation

The interpretation is correct, and the writeup gets a subtle point right that the catalog got wrong.

c_r is defined in the writeup as $c_r = \inf_{\text{partitions } N} \max_i d_{\log}(\Sigma(A_i))$. I fetched the source paper (arXiv:2105.15195v3, downloaded and converted to text) and this is verbatim CFP's definition ("let c_r be the minimum, taken over all partitions of N into r parts, of the maximum over i of the upper logarithmic density of $\Sigma(A_i)$ "). $\Sigma(A)$ is sums of distinct elements, $d_{\log}$ is $\limsup_x (1/\log x) \Sigma_{\{n \leq x, n \in S\}} 1/n$. Both match.

The catalog's $2^r b_0 - r$ is a PDF-extraction garble. CFP's Theorem 1 reads $(1 - 1/(2b_0))(1 + 1/(2rb_0 - r))$ with b_0 the root > 1 of $b^r - 2rb + r - 1$. The writeup's §6 diagnoses this correctly and proves the intended $2r$ version; it also correctly observes the two coincide at $r = 2$, which is why the $r=2$ sanity check hides the error. This is exactly right and is confirmed against the source PDF (line 119 ff. of my extraction). Marking the literal 2^r reading as "false even as an upper bound" is a fair call, not interpretation gaming: a reasonable author would regard the $2r$ statement as the conjecture.

So the target is CFP's **Conjecture 10** (§4 of the source paper): $c_r = b_0^{r-1}/(2r)$ for $r \geq 3$. The writeup claims to prove it for all $r \geq 2$. That is the intended problem.

What the writeup gets wrong is not the statement but the diagnosis of what is open. The writeup asserts (intro and §6) that "the key addition ... is an additive lemma for dense sets of primes. It provides the required intervals of subset sums without requiring any monochromatic interval of integers", and closes with "the gap in the previous attempt is closed by Lemma 3". That step was never the gap — see the Reference check.

B.2 Step-by-step findings

#	step	label	note
1	Theorem, constant identity (1): $b^{r-1}/(2r) = (1 - 1/(2b))(1 + 1/(2rb - r))$	VALID	Proved symbolically: (lhs-rhs) mod $(b^r - 2rb + r - 1) = 0$ for $r=2..8$ (sympy), and numerically to 12 digits for $r=2..10$. $r=2$ gives $b=2+\sqrt{3}$, $c_2=(2+\sqrt{3})/4$.
2	§1 def. of $D_{\{r,\lambda\}}$, $C_{\{r,\lambda\}}$, and 0 $< C_{\{r,\lambda\}} < 1$	VALID	$D_{\{r,\lambda\}}(q) > 1 - 1/(\lambda q) > 1 - 1/\lambda$; < 1 at the minimiser since $b_{\{r,\lambda\}}^{r-1} > \lambda$.
3	Lemma 1, component structure (C $= [t_f, \lambda t_{\ell+1}]$, no unbounded component)	VALID	An unbounded component contains a ray $\Rightarrow$ density $1 > D$. Only finitely many j_k meet a bounded set ($t_k \rightarrow \infty$), so each component is a closed interval whose endpoints are the extreme indices. Verified by brute force on explicit configurations: 0 structure failures in all 22 tested configurations (script 07).
4	Lemma 1, recurrence (6) $d_C = a_C + d_{\{C^-\}}/R_C$	VALID	Exact; U_i has no measure between $e_{\{C^-\}}$ and t_f . Checked numerically to $1e-9$ on all components of all test configurations.
5	Lemma 1, (7) $1 < Q_C < Q_{\max} = 1/(\lambda(1-D))$	VALID	$a_C < d_C \leq D$ and $a_C = 1 - 1/(\lambda Q_C)$ give it directly; $Q_C > 1$ since $\ell+1 > f$.
6	Lemma 1, (8)	VALID	$d_{\{C^-\}}/d_C = R_C(1 - a_C/d_C)$ is an identity; $d_C \leq D$ gives $1 - a_C/d_C \leq 1 - a_C/D$. Identity checked numerically.
7	Lemma 1, uniform η from $F(q) = \sqrt{r} \log q + \log(1 - 1/(\lambda q))/D$ (9)	VALID	$F \geq 0 \Leftrightarrow D \geq D_{\{r,\lambda\}}(q)$, impossible since $D < C_{\{r,\lambda\}} = \inf D_{\{r,\lambda\}}$. $F(1) = \log(1 - (1 - 1/\lambda)/D) < 0$, $F \rightarrow -\infty$ at $Q_{\max}$, so $\sup_{[1, Q_{\max})} F$ is attained on a compact subinterval and is < 0 .

#	step	label	note
8	Lemma 1, (10) LHS telescopes, $O(1)$	VALID	Retained components of a colour with $\ell \leq N$ form an initial segment; $\text{sum} = \log(d_{\text{first}}/d_{\text{last}})$ with $1 - 1/\lambda \leq d \leq 1$. Needs a <i>lower</i> bound on the LHS, which is what boundedness supplies — correct.
9	Lemma 1, (11) $\Sigma \log R_C \leq r \log t_{\{N+1\}} + O(1)$	VALID	Per colour telescopes to $\log(e_{\text{last}}/e_{\text{first}})$, $e_{\text{last}} \leq \lambda t_{\{N+1\}}$, e_{first} a fixed constant. Numerically $\Sigma \log R - r \log t_K$ stays bounded above (values $-7 \dots -33$, no upward drift) over K from 120 to 128000 (script 07).
10	Lemma 1, (12) covering estimate $\Sigma \log Q_C \geq \log t_{\{N+1\}} - O(1)$	VALID	Each index lies in the span of its own colour's component; discarded components and the $\leq r$ components straddling N contribute $\leq r \log Q_{\text{max}} = O(1)$. Numerically $\Sigma \log Q - \log t_K$ stays bounded below ($-2.4 \dots -4.8$, and positive when spans overlap) with no downward drift.
11	Lemma 1, contradiction ($m_N \rightarrow \infty$)	VALID	The $r \log t_{\{N+1\}}$ from (11) exactly cancels $r(12)$, leaving $O(1) \leq O(1) - \eta m_N$. Infinitely many components since none is unbounded and each holds finitely many j_k . I re-derived the whole lemma independently and it holds.
12	(13) $D'_{\{r,\lambda\}}(q)=0 \iff q^r - \lambda r q + r - 1 = 0$	VALID	sympy: numerator of D' equals $q^{r-2} \cdot (q^r - \lambda r q + r - 1)$ exactly, for $r=2..7$.
13	(14) $C_{\{r,\lambda\}} = b_{\{r,\lambda\}}^{r-1}/(\lambda r)$, $b_{\{r,\lambda\}} > \lambda^{1/(r-1)}$, (15) continuity as $\lambda \uparrow 2$	VALID	$b^{r-1} = r(\lambda b - 1)$ gives (14) by hand; $P(1)=r(1-\lambda)<0$ and P decreasing on $(1, \lambda^{1/(r-1)})$ gives the root location. Numerically $C_{\{r,\lambda\}} \rightarrow C_{\{r,2\}}$ ($r=2$: .9251, .93228, .93294 at $\lambda=1.9, 1.99, 1.999$).
14	Lemma 2 statement ($k(s-k) \geq p-1 \implies \forall \text{element of } F_p \text{ is a sum of } k \text{ distinct members of } B$)	VALID (known)	This is the Dias da Silva–Hamidoune theorem $ k \wedge B \geq \min(p, k(s-k)+1)$. The writeup flags it as “standard”, which is honest. Brute-forced: 7740 (p,B,k) cases over $p \in \{5, \dots, 37\}$, 2076 satisfying the hypothesis, 0 failures , and the DdSH bound itself never violated.
15	Lemma 2 proof: P vanishes on B^k , choice of d_i , degree bookkeeping	VALID	The explicit d_i ($p-1 = kq+u$; $q, \dots, q+k-u-1$ then $q+k-u+1, \dots, q+k$) is distinct, increasing, $\geq 0, \leq s-1$, and sums to $p-1+C(k,2)$ — brute-checked over all (p,s,k) with $k(s-k) \geq p-1$, p up to 211, s up to 59: 0 failures . The range claim “sums of k distinct elements of $\{0, \dots, s-1\}$ fill $[C(k,2), C(k,2)+k(s-k)]$ ” verified exactly for $s \leq 12$.

#	step	label	note
16	Lemma 2, coefficient formula (17) $= (p-1)! / \prod d_i! \cdot \prod_{\{i < j\}} (d_j - d_i)$	GAP (routine)	Asserted “as follows by expanding the Vandermonde determinant”. The required identity is $\det(1/(d_i - j + 1)!)_{\{i,j\}} = \prod_{\{i < j\}} (d_j - d_i) / \prod_i d_i!$; it is <i>not</i> proved. It is true (multiply row i by $d_i!$ to get falling factorials $(d_i)_{\{j-1\}}$, monic of degree $j-1$, so the determinant is the Vandermonde $\prod_{\{i < j\}} (d_j - d_i)$). I verified it; it is a routine repairable omission. Non-vanishing mod p is fine since $d_i \leq s-1 \leq p-1$ and $0 < d_j - d_i < p$.
17	Lemma 3 Step 1 (groups summing to 0 mod p , $d = \ell p$, $L \geq m/(8h^2)$, $T \geq mn/(8h)$)	VALID	$h(v-h) \geq h \cdot m/8 \geq p$ for $v \geq m/4$; group sum $\in [hn, 2hn]$ and divisible by p gives $h/4 \leq \ell \leq h$; $d \geq hn$ holds because d is the group sum. Number of groups $\geq \lfloor (m/2) \rfloor \setminus \lfloor m/4 \rfloor / h \geq m/(8h)$ (the writeup’s $m/(8h)$ is conservative; the true count is $\approx m/(4h)$). $T \gg \kappa n^2/(\log n)^2$ follows.
18	Lemma 3 Step 2 (reserve set, $\gcd(c, \ell)=1$, CRT, $M = 2n(H+\ell)$)	VALID	This is the only place primality is used, and it is used correctly: every $a \in A^{\wedge\{1\}}$ is a prime $> \ell$, so $\gcd(a, \ell)=1$, so the chosen residue class c is a unit mod ℓ and $t \mapsto tc$ sweeps all residues mod ℓ over ℓ consecutive t . $t(s-t) \geq H \cdot s/2 \geq p$ needs $H+\ell \leq s/2$, which holds for large n . $\gcd(p, \ell)=1$ since $\ell \leq h < p$.
19	Lemma 3 Step 3 (fill $[M, T]$, disjointness)	VALID	$x \geq M \geq b_x$ and $x - b_x \leq T = Ld$ give $0 \leq q \leq L$; the groups lie in $A^{\wedge\{0\}}$, the reserve subset in $A^{\wedge\{1\}}$, so they are disjoint. $M \setminus \setminus O_{\kappa}(n(\log n)^2) \setminus \leq n(\log n)^3$ and
20	Lemma 3 as a whole	VALID but REDUNDANT	$T \gg \kappa n^2/(\log n)^2 \geq n^2/(\log n)^3$. Strictly weaker than CFP’s own Lemma 2 (source paper §2), which is already published — see Reference check.
21	§4 reduction (PNT pigeonhole, $t_k \setminus \setminus \log n_k \setminus + 3 \log \log n_k$, (24), (25), $U_i \subseteq W_i$)	VALID	$(2-\lambda)k \log 2$ dominates the $O(\log k)$ remainder, so $\lambda t_{\{k+1\}} \leq 2 \log n_k - 3 \log \log n_k$ for large k . The $U_i \setminus W_i$ bound $\sum_k e^{\{-t_k\}} < \infty$ is correct (flooring loses $\leq \log(1+e^{\{-t_k\}})$ at the left end of each J_k , and $t_k \sim k \log 2$). $ W_i \cap [0, T] \leq \sum_{\{n \leq e^T, n \in E_i\}} \log(1+1/n) \leq \sum 1/n$ is the right direction.
22	§4 conclusion $c_r \geq b^{\wedge\{r-1\}}/(2r)$	VALID	Follows from Lemma 1 + (15), $\lambda \uparrow 2$. But this whole section is CFP’s §3 re-derived.
23	§5 matching upper bound	VALID but already published	$\Sigma(A_i) \subseteq U_{\{k \equiv i\}}[N_k, N_{\{k+1\}}^2]$ is correct ($\sum_{\{n < N\}} n = N(N-1)/2 < N^2$); disjointness from $b > 2^{\wedge\{1/(r-1)\}}$; the density limit (29) recomputed independently as $(2b-1)/(2b(1-b^{\wedge\{-r\}})) = C_r$. “Density increases inside an occupied interval and decreases in a gap” is correct ($d/dT(m/T) > 0 \iff m < T$). This is CFP’s Theorem 1.

#	step	label	note
24	§6 2^r vs $2r$	VALID	Confirmed against the source PDF. $2^r > 2r$ for $r \geq 3$, so the literal catalog formula is strictly below c_r .

No step is an ERROR. The only genuine mathematical omission is #16, which is routine.

B.3 Reference check

I downloaded and read the full source paper (<https://arxiv.org/pdf/2105.15195v3>). The writeup cites **nothing at all**, and the reference check is where it comes apart.

1. **CFP's Lemma 2 already is the writeup's Lemma 3, in stronger form.** Source paper §2:

Lemma 2. For every positive integer r , there are positive constants $C = C(r)$ and $C' = C'(r)$ such that the following holds. For every $N > 0$ and every partition $N \cap [N, eN] = A_1 \sqcup \dots \sqcup A_r$ into r colour classes, there is some $i \in [r]$ such that $\Sigma(A_i)$ contains all positive integers in $[CN, C'N^2]$.

That is precisely “at every scale, one colour’s subset sums fill an interval from $\sim N$ to $\sim N^2$, with no monochromatic interval of integers required” — the exact thing the writeup advertises as its “key addition”. CFP’s version is **stronger**: the endpoints are CN and $C'N^2$ with constants, whereas the writeup loses $(\log n)^{\pm 3}$ at both ends. CFP prove it from Theorem 6.1 of their companion paper arXiv:2104.14766 (with an alternative route via Szemerédi–Vu, Remark 5), using integers free of prime factors $\leq r^2$ instead of primes — the same “no small common divisor” trick, implemented better. The writeup’s §4 (auxiliary colouring by dyadic scale, transfer to logarithmic density) is CFP’s §3 and their Lemma 6.

2. **The actual open step is the writeup's Lemma 1, and the writeup never says so.** CFP’s Lemma 7 is *exactly* the writeup’s Lemma 1 specialised to $r = 2$, $\lambda = 2$: they prove that $S(\varphi, i) = \cup_{\{ \alpha(j)=i \} [j, 2j]}$ has upper density at least $f_2 = \inf_{\{z \in [0, 1]\}} (1 - z/2)/(1 - z^2)$, which is $C_{\{2, 2\}}$ under $z = 1/q$. They prove it with a Brouwer fixed-point argument on the pair $(\bar{a}_n, \bar{b}_n)$ (source paper §3, Claims 8 and 9), and their concluding remarks say:

“We have explored this conjecture in some detail ourselves, but were unable to establish the optimality of our upper bound for c_r without additional assumptions. For instance, it seems that our methods do apply if the auxiliary colouring α defined in Section 3 is assumed to be **cyclic** ... Since every two-colouring of the positive integers is automatically cyclic in this sense, this restriction does not hamper us in that case.”

So the obstruction is entirely in the geometric lemma for non-cyclic colour words, and the writeup’s Lemma 1 is a direct attack on it. The writeup does not mention Lemma 7, Brouwer, the cyclic restriction, or Conjecture 10 by name.

3. **Inheritance from the previous attempt.** Lemma 1 is *not new in this writeup*: it is the previous model’s Theorem 2.1, with the same construction and the same telescoping/covering skeleton. The retry tightened the boundary accounting in (12) and the derivation of η , and dropped the previous attempt’s unnecessary reduction to $s \leq r$ active colours. So the load-bearing novelty is inherited, and the part the writeup advertises as its contribution is the redundant one. Both models misdiagnosed which half of CFP’s argument was missing: the previous attempt declared the additive reduction to be the gap (it is not — CFP did it in 2021) and reported partial; the retry “closed” that non-gap and reported proved.
4. Externally invoked results are all stated correctly: **Bertrand’s postulate** (a prime in $(2n, 4n)$ — apply Bertrand to $2n$; valid for every $n \geq 1$), the **prime number theorem** ($\pi(2n) - \pi(n) \geq n/(2 \log n)$ for large n), **Fermat’s little theorem**, and the **Combinatorial Nullstellensatz** (the coefficient form, sketched via Lagrange interpolation). Lemma 2 is the Dias da Silva–Hamidoune theorem (1994), whose polynomial proof is due to Alon–Nathanson–Ruzsa; the writeup labels it “standard” but names nobody.
5. **Priority (step 2b).** Nothing found. Searches over the citing literature and 2025–2026 arXiv listings turned up no paper resolving Conjecture 10 for $r \geq 3$; the catalog’s own May-2026 review reports the same. The companion paper arXiv:2104.14766 (to appear in JEMS) develops the additive machinery but does not settle the tightness conjecture. So the *theorem* is not already known — but the writeup’s self-declared key lemma is, and that is why I set `references_ok: false`.
6. **No self-contamination (step 2c).** The prior art above is not a web paraphrase. I downloaded <https://arxiv.org/pdf/2105.15195v3> directly, converted it with `pdftotext`, and read Lemma 2, Lemma 7 and the concluding remarks in the original text; the paper has named authors (Conlon, Fox, Pham), an arXiv identifier, a v3 date of 22 Sep 2022 and a venue (Mathematika 68 (2022), 1292–1301). It predates this campaign by four years and is the very paper the prompt links as the source. Nothing I recorded as known traces back to this repository.

B.4 Computational check

Scripts in `verification_astra/scripts/2105.15195__00/` (run with `system python3`; the repo `.venv` has no `numpy`). 01, 03, 04, 05 survive from the killed earlier run and were re-run; 06–09 are new. Note that the earlier run’s `02_geometric_lemma.py / geom_out.txt` reported a **spurious** “COUNTEREXAMPLE” at density 0.5 — that script truncated the configuration at 60 periods, so configurations whose breakpoint ratios tend to 1 (where every interval in fact merges into one *unbounded* component of density 1) were scored as 1/2. 06/08 replace it with an exact self-similar limsup and get this right.

- **01_constants.py** — b_0 and $C_r = b_0^{\frac{r-1}{2r}}$ agree with $\inf_q D_{\{r,2\}}(q)$ and with $(1 - \frac{1}{2b})(1 + \frac{1}{2rb - r})$ to 12 decimals for $r = 2..10$ ($r=2$: $0.933012701892 = (2 + \sqrt{3})/4$; $r=3$: 0.852624899444 , $b_0 = 2.261802245260$; $r=4$: 0.797839606231 ; $r=5$: 0.759010545363). The literal 2^r formula gives 0.830541680722 ($r=3$), 0.758897892478 ($r=4$) — strictly smaller, confirming §6. $C_{\{r,\lambda\}} \rightarrow C_{\{r,2\}}$ as $\lambda \uparrow 2$ confirmed.
 - **03_lemma2_ddsh.py** — Lemma 2 brute force: **7740** (p, B, k) cases, **2076** with $k(s-k) \geq p-1$, **0 failures**; the Dias da Silva–Hamidoune bound $|k \wedge B| \geq \min(p, k(s-k)+1)$ never violated; the interval-of-sums claim and the writeup’s explicit d_i choice verified over all valid (p,s,k) with $p \leq 211$, $s \leq 59$ — **0 failures**.
 - **04_prime_blocks.py** — Lemma 3’s claimed interval $[n(\log n)^3, n^2/(\log n)^3]$ is **empty for $n < 2.4 \times 10^7$** , and the proof’s internal inequalities ($h \leq m/8$, $H + l \leq s/2$) only all hold from about $n \approx 10^{15}$ ($r=2,3$). Exact bitset DP on real prime sets corroborates the qualitative content: for $A = a$ κ -fraction of the primes in $[n, 2n]$, $\Sigma(A)$ contains a single run of consecutive integers from $\approx 4n$ to $\approx 0.18 n^2$ ($n = 3000$, $\kappa = 0.94$: $[12101, 1568810]$; $\kappa = 0.235$: $[13183, 394161]$).
 - **05_construction.py** — the upper-density of $V_i = \cup_{k \equiv i (r)} [b^k, 2b^{k+1}]$ measured directly equals C_r to 12 decimals for $r = 2..8$, the intervals are disjoint, and $b_0 > 2^{\frac{1}{r-1}}$ holds throughout.
 - **07_lemma1_mechanics.py** — the structural claims inside Lemma 1’s proof, on explicit (t_k, γ_k) : every component is exactly $[t_k, \lambda t_{k+1}]$, the recurrence (6) and the identity behind (8) hold to $1e-9$, and across $K = 120 \dots 128000$, $\Sigma \log R_C - \log t_K$ stays bounded above and $\Sigma \log Q_C - \log t_K$ stays bounded below, with no drift — i.e.
- (11) and (12) are genuine $O(1)$ bounds. **0 failures in 22 configurations** (extremal cyclic, arithmetic $t_k = k$ with random non-cyclic runs, length-1 runs).
- **08_geometric_search.py** (with `06_geometric_exact.py`) — the decisive test. Exact limsup for log-periodic configurations, minimised over log-gaps by multi-start Nelder–Mead, for **every colour word up to rotation and colour permutation** (so all non-cyclic patterns CFP cannot handle are included): $r=2$ up to $P=8$, $r=3$ up to $P=7$, $r=4$ and $r=5$ up to $P=6$. **No configuration beats C_r** . The minimum is attained exactly at the cyclic equal-ratio word with run ratio b_0 ($r=3$: 0.852624899444 at ratios all 2.2618 ; $r=4$: 0.797839606231 at 1.85496 ; $r=5$: 0.759010545363 at 1.65982), and the best non-cyclic words come in strictly above ($r=3$, word $[0,1,0,2]$: $0.870233 = C_3 + 1.8e-2$; $r=4$, word $[0,1,0,2,3]$: 0.826365 ; $r=5$, word $[0,1,2,0,3,4]$: 0.780120). Since any geometric configuration can be realised by an actual block colouring of N , a value below C_r would have disproved CFP’s Conjecture 10 outright; none exists in this range.
 - **09_symbolic.py** — sympy: numerator of $D'_{\{r,\lambda\}}$ equals $q^{\frac{r-2}{2}}(q^r - \lambda r q + r - 1)$ exactly ($r=2..7$), and $b^{\frac{r-1}{2r}} - (1 - \frac{1}{2b})(1 + \frac{1}{2rb - r}) \equiv 0 \pmod{(b^r - 2rb + r - 1)}$ ($r=2..8$).

Nothing failed computationally.

B.5 Caveats

1. **Attribution / novelty (the main one).** Lemma 3 duplicates, more weakly, CFP’s published Lemma 2; §4 duplicates their §3 and Lemma 6; §5 duplicates their Theorem 1. The one step that is actually new — Lemma 1 for $r \geq 3$ non-cyclic colour words — was inherited from the previous attempt and is nowhere flagged as the contribution. As written the paper would be rejected on scholarship grounds even if every line is right.
2. **Omitted determinant identity** in Lemma 2 (step #16). Routine; I checked it.
3. **Lemma 3 has no finite content:** the interval it produces is empty below $n \approx 2.4 \times 10^7$ and the proof’s parameter inequalities only bite around $n \approx 10^{15}$. This is legitimate (everything is “for sufficiently large n ”) but it means no computational check can validate Lemma 3 directly; I could only verify its internal inequalities symbolically and its qualitative content on small dense prime sets.
4. **Prime-existence risk (flagged in my brief): clean.** Bertrand is used only to pick a prime in $(2n, 4n)$, which exists for every $n \geq 1$ (apply Bertrand to $2n$); no stronger prime-gap input is needed and nothing depends on a prime existing in a shrinking window. The PNT is used only as a lower bound $\geq n/(2 \log n)$ for large n . Every use is asymptotic, so there is no sub-threshold small-case obligation.
5. **Interlacing risk: not applicable.** “Interlacing” here means interlacing of the colour classes along N , not eigenvalue interlacing. There is no matrix, no submatrix, and no spectral argument anywhere in the writeup, so the principal-submatrix / symmetry failure mode cannot arise.
6. **$r = 1$ is excluded** (the polynomial has no root > 1); the theorem is stated for $r \geq 2$, which is the right range. Finitely many integers may be recoloured arbitrarily throughout — harmless for a limsup density, and correctly noted in §5.
7. **$\lambda \uparrow 2$ is a genuine limit, not an attained value.** For each $\lambda < 2$ the threshold index $k_0(\lambda)$ differs; this is fine because the conclusion is a limsup, but it means the proof never produces the intervals $[t_k, 2t_{k+1}]$ directly. The writeup handles it correctly (the $(2-\lambda)k \log 2$ slack absorbs the $3 \log \log n$ erosion).
8. **would_publish: true in the verdict block is over-claimed** relative to what the document is: the mathematics may well be publishable, the document is not.
9. My confidence is medium, not high, deliberately. I re-derived Lemma 1 from scratch line by line, verified its structural claims numerically, and found no counterexample to its conclusion in an exhaustive word search — but it is a short elementary argument settling a point that Conlon, Fox and Pham state they “explored in some detail” without success, and a single referee pass is not the same as the scrutiny such a claim deserves. The residual risk is concentrated entirely in Lemma 1’s $O(1)$ bookkeeping (steps 8–11).

B.6 Referee summary

The writeup claims, and as far as I can verify actually proves, Conlon–Fox–Pham’s Conjecture 10: $c_r = b_0^{\frac{r-1}{2r}}$ for every $r \geq 2$, with b_0 the root > 1 of $b^r - 2rb + r - 1$. Every step checks out: the constant identities are confirmed symbolically, Lemma 2 is the Dias da Silva–Hamidoune theorem and passes 7740 brute-force cases, Lemma 3’s parameter chain is sound, the reduction in §4 and the construction in §5 are correct, and the geometric Lemma 1 — the only thing here that is not already in the literature — survives an independent re-derivation, a numerical audit of its telescoping and covering estimates, and an exhaustive search over all non-cyclic colour words for $r \leq 5$ that found no configuration below C_r . The single mathematical omission is the Vandermonde determinant identity behind the Nullstellensatz coefficient (17), which is routine and which I verified. What sinks the presentation, and what stops this being CONFIRMED, is that the writeup misidentifies its own contribution: its advertised “key addition”, the prime-based Lemma 3 that supposedly “closes the interlacing gap”, is a strictly weaker re-derivation of CFP’s own Lemma 2 from the very paper the prompt links, and the interlacing step was never the gap — CFP

settled it in 2021 and said so. The real obstruction, which CFP’s concluding remarks name explicitly (their Lemma 7 holds for $r = 2$ by a Brouwer argument, and for $r \geq 3$ only under a cyclic-colouring assumption), is the writeup’s Lemma 1, which it inherited from the previous attempt and never identifies as the novel step. MINOR_GAPS at medium confidence: correct modulo one routine omission, but requiring a complete rewrite of its framing and a careful human check of Lemma 1’s $O(1)$ bookkeeping before anyone should believe a forty-year-old Erdős-lineage conjecture has fallen to a page and a half of telescoping.

References

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- [2] D. Conlon, J. Fox and H. T. Pham, *Subset sums, completeness and colorings*, [arXiv:2104.14766](#).
- [3] J. A. Dias da Silva and Y. O. Hamidoune, *Cyclic spaces for Grassmann derivatives and additive theory*, *Bulletin of the London Mathematical Society* **26** (1994), 140–146.
- [4] N. Alon, M. B. Nathanson and I. Z. Ruzsa, *The polynomial method and restricted sums of congruence classes*, *Journal of Number Theory* **56** (1996), 404–417.